NavEye: An Edge-Native, Confidence-Gated Multi-Modal Vision and Spatial Guidance Framework for the Visually Impaired

Sanjay P
(310122104085)
Dept. of Computer Science & Eng.
Anand Institute of Higher Tech.
Chennai, India
sanjayp.cse@aiht.ac.in

Punuja Lokith S
(310122104075)
Dept. of Computer Science & Eng.
Anand Institute of Higher Tech.
Chennai, India
lokiths.cse@aiht.ac.in

Mrs. R. Pratheeba
Supervisor
Dept. of Computer Science & Eng.
Anand Institute of Higher Tech.
Chennai, India
pratheebar.cse@aiht.ac.in

Mrs. M. Maheswari
Head of Department
Dept. of Computer Science & Eng.
Anand Institute of Higher Tech.
Chennai, India
maheswarim.cse@aiht.ac.in

***Abstract*—**Visual impairment affects over 43 million individuals worldwide, severely restricting independent mobility, spatial orientation, and daily social functioning. While the traditional white cane detects ground-level obstacles, it remains blind to elevated hazards, overhead barriers, and contextual environmental semantics. Modern smartphone-based assistive technologies attempt to bridge this divide using deep neural vision; however, existing cloud-reliant systems exhibit severe latency spikes (1.2–3.5 s), fail during network dropouts, and deliver verbose conversational descriptions that induce acoustic fatigue rather than providing immediate, actionable steering directives. We present NavEye, a comprehensive edge-native, real-time multi-modal assistive navigation framework running entirely on resource-constrained mobile hardware. NavEye couples an INT8-quantized YOLOv8n object detector operating at 28.4 frames per second with a monocular pinhole geometric depth engine to continuously estimate obstacle distance. A 3-corridor horizontal spatial reasoning algorithm with boundary hysteresis (guard band $\delta = 0.04$) eliminates steering vector oscillations. To guarantee human safety, a confidence-gated priority preemption queue enforces a 'Never Silence' policy, arbitrating between sub-15 ms emergency collision stops (<0.8 m), calibrated evasive maneuvers, and low-latency bilingual audio synthesis in English and native Tamil (ta-IN). Furthermore, the framework integrates offline Indian currency identification and privacy-preserving ArcFace metric face verification. Extensive empirical evaluation across 500 multi-environment indoor and outdoor obstacle trials proves that NavEye achieves a 95.8% collision avoidance rate with a deterministic 41.6 ms end-to-end guidance latency, reducing user cognitive workload by 61.4% compared to standard cloud-assisted alternatives.

***Index Terms*—**assistive computer vision, edge AI, monocular depth estimation, spatial corridor reasoning, visually impaired navigation, bilingual voice guidance, YOLOv8, MobileFaceNet, hysteresis.

I. INTRODUCTION

Autonomous and safe physical navigation within dynamically changing, unstructured environments represents one of the most formidable daily challenges encountered by blind and visually impaired (BVI) persons. According to epidemiological data from the World Health Organization (WHO), approximately 43 million people suffer from complete blindness, with over 295 million living with moderate-to-severe visual impairment. In developing nations such as India, visually impaired citizens frequently traverse dense urban corridors, irregular footpaths, chaotic vendor stalls, low-hanging tree branches, and construction obstacles that lack tactile paving or universal accessibility infrastructure [10]. For more than a century, the primary mobility tool for the blind has been the traditional mechanical white cane. Although mechanically robust and intuitive for ground tactile exploration, the white cane suffers from fundamental physical limitations: its detection range is restricted to approximately one meter at foot level, leaving the user critically vulnerable to head-height, chest-level, or rapidly approaching overhanging obstacles.

Recent advances in ubiquitous mobile computing and consumer smart devices have catalyzed significant research into camera-based Electronic Travel Aids (ETAs). Because modern smartphones incorporate high-resolution image sensors, Inertial Measurement Units (IMUs), and embedded Neural Processing Units (NPUs), they represent a highly cost-effective, non-stigmatizing platform capable of hosting advanced computer vision solutions. A multitude of commercial applications—such as Microsoft Seeing AI, Envision, and Be My Eyes—have demonstrated the utility of deep learning for object classification, text reading, and scene understanding. Nevertheless, when evaluated against the stringent requirements of real-time safe pedestrian navigation, contemporary solutions suffer from three critical systemic deficiencies:

1) Cloud Latency and Connectivity Failure: Most state-of-the-art vision systems offload image frames to remote cloud servers for large vision-language model inference. Under real-world cellular network conditions, network round-trip latency fluctuates widely between 1,200 ms and 4,000 ms. At an average human walking velocity of 1.2 m/s, a 2.5-second processing delay translates to 3.0 meters of blind forward travel, rendering collision detection dangerously obsolete before the warning is spoken.

2) Lack of Actionable Spatial Steering: Generic computer vision tools merely enunciate isolated semantic labels (e.g., 'chair', 'motorcycle') without computing relative geometric distance, heading trajectories, or lateral corridor occupancy. Without knowing whether a detected obstacle impedes their immediate walking vector, visually impaired users cannot formulate decisive evasive steering actions.

3) Acoustic and Linguistic Exclusion: Existing assistive tools generate lengthy, uncalibrated natural language sentences that mask crucial environmental audio cues (such as approaching vehicles, footsteps, or echoes). Furthermore, the overwhelming majority of commercial solutions restrict

speech synthesis to English, fundamentally alienating non-English-speaking demographics across regional Indian states such as Tamil Nadu [9].

To overcome these fundamental barriers, we introduce NavEye, a holistically engineered, edge-native assistive navigation framework that executes all sensor perception, geometric distance estimation, corridor spatial reasoning, and voice arbitration entirely on consumer mobile hardware. Specifically, the major technical contributions of this work are summarized as follows:

- **An edge-native, zero-cloud perception pipeline:** We implement an INT8-quantized YOLOv8n detector integrated with a lightweight MobileFaceNet ArcFace model, executing asynchronously in dedicated background threads at 28.4 FPS on mobile NPUs with deterministic sub-42 ms latency.
- **A 3-corridor spatial reasoning engine with boundary hysteresis:** We formulate a horizontal Field-of-View (FoV) partition into Left, Center, and Right zones, governed by a boundary hysteresis band ($\delta = 0.04$) and monocular pinhole distance estimation that eliminates oscillating steering alerts.
- **A confidence-gated priority preemption queue:** We establish an absolute safety-critical arbitration state machine enforcing a 'Never Silence' policy, which guarantees immediate acoustic override (<15 ms) for critical collision risks within 0.8 m while suppressing duplicate redundant notifications.
- **A low-latency bilingual directive synthesizer:** We design an actionable audio engine delivering synchronized Tamil (ta-IN) and English voice guidance and directional haptic patterns, minimizing cognitive auditory load.
- **Rigorous empirical and usability validation:** We conduct 500 real-world obstacle trials across varied indoor/outdoor scenarios and a human usability trial with visually impaired participants, proving a 95.8% collision avoidance rate.

The remainder of this paper is structured as follows. Section II reviews related literature in electronic travel aids, edge vision, and depth modeling. Section III formalizes the mathematical problem. Section IV details the proposed NavEye system architecture. Section V presents implementation specifications. Section VI discusses experimental benchmarks and ablation studies. Section VII analyzes safety ethics and limitations, and Section VIII concludes with future directions.

II. RELATED WORK

A. Electronic Travel Aids and Wearable Sensors

Early electronic travel aids relied heavily on specialized ultrasonic, infrared, or LiDAR range-finders mounted onto canes, glasses, or waist belts [3], [8]. While active sensor bands deliver accurate range measurements, they impose severe physical burdens: bulky battery packs, fragile sensor arrays, and high manufacturing costs ($800–$3,500). Furthermore, distance sensors lack semantic scene understanding; an ultrasonic transducer cannot distinguish between an impending stairwell, an empty chair, or an approaching human guide. In contrast, monocular smartphone vision delivers rich semantic perception without supplementary hardware [4].

B. Real-Time Deep Learning Object Detection on Edge Devices

Convolutional object detectors have evolved rapidly from two-stage architectures (Faster R-CNN) to single-stage real-time models (SSD, YOLO) [1]. Standard models require substantial GPU resources incompatible with mobile hardware. While MobileNet-SSD provided early mobile feasibility, it suffered from low mean Average Precision (mAP) on small, low-contrast obstacles. Recent advancements in YOLOv8 and YOLOv11 incorporate anchor-free decoupled heads and Cross-Stage Partial architectures, yielding superior feature representation [11]. By applying post-training INT8 quantization via ONNX Runtime, we achieve near-lossless detection accuracy while slashing inference latency down to 28.4 ms on mobile processors.

C. Monocular Depth Estimation and Obstacle Avoidance

Stereoscopic vision and monocular deep learning depth estimators (e.g., MiDaS, Depth-Anything) generate dense pixel-wise depth maps but require billions of floating-point operations per frame, bottlenecking mobile framerates to below 4 FPS [6], [12]. In contrast, pinhole geometric projection models utilize real-world physical object height priors to derive reliable distance estimates in sub-millisecond runtimes. However, unconstrained phone tilting introduces perspective foreshortening errors. NavEye resolves this by fusing on-board IMU pitch telemetry into a perspective-corrected pinhole formulation.

D. Assistive Identity, Currency, and Ambient Perception

Social inclusion requires more than physical obstacle avoidance; visually impaired individuals frequently report severe anxiety surrounding personal identification and financial transactions. Traditional FaceNet models utilize heavy ResNet backbones unsuitable for on-device continuous tracking. MobileFaceNet employs depthwise separable convolutions optimized by ArcFace angular margin loss [2], [9]. We extend this by integrating multi-angle facial template averaging and localized Indian currency recognition into an integrated multi-modal framework.

E. Positioning of Proposed Framework

Unlike prior systems that either isolate object detection without spatial steering or rely on cloud connectivity, NavEye is built upon three architectural cornerstones: (i) 100% on-device deterministic execution, (ii) boundary hysteresis-guarded spatial corridor reasoning, and (iii) safety-first priority preemption with bilingual Tamil/English audio guidance.

III. PROBLEM FORMULATION

Let a continuous sensor frame stream captured at discrete time steps t be denoted by $F_t \in \mathbb{R}^{(H \times W \times 3)}$, accompanied by IMU pitch angles ϕ_t. The visual perception pipeline maps F_t to a set of k detected bounding boxes $B_t = \{b_1, ..., b_k\}$, where each obstacle tuple b_i is defined as $b_i = (x_i, y_i, w_i, h_i, c_i, s_i)$, corresponding to normalized center coordinates, dimensions, class label, and detector confidence.

The horizontal Field-of-View is partitioned into three mutually disjoint spatial corridors along normalized width x

∈ [0, 1]: Left corridor Ω_L = [0, θ_L), Center path Ω_C = [θ_L, θ_R], and Right corridor Ω_R = (θ_R, 1], where nominal boundaries are θ_L = 0.35 and θ_R = 0.65. Let D_i(t) denote the estimated distance to obstacle b_i. The dynamic collision risk index R_i(t) is modeled as a multi-factor continuous function:

$$R_i(t) = \gamma_1 / \max(D_i(t), \varepsilon) + \gamma_2 \cdot \mathbb{I}(x_i \in \Omega_C) + \gamma_3 \cdot (dD_i / dt) \quad (1)$$

where $\mathbb{I}(\cdot)$ is the indicator function, and γ1, γ2, γ3 are positive weighting coefficients. The guidance objective is to learn an arbitration policy π*(Bt, {Ri}) that maps environmental state to an actionable acoustic directive at ∈ {STOP, STEER_L, STEER_R, HEARTBEAT, IDLE} such that collision probability P(collision) is minimized while directive issuance frequency remains below the human cognitive overload threshold.

IV. PROPOSED SYSTEM ARCHITECTURE

A. Multi-Modal Ingestion and Edge Preprocessing

The system pipeline is structured as an asynchronous multi-threaded state machine, illustrated in Fig. 1. High-resolution YUV420 camera frames are ingested from Android CameraX at 30 FPS into a background Dart isolate. Frames are zero-copy converted to planar RGB buffers and downsampled to 320×320 (for obstacle detection) and 112×112 (for face verification). Concurrently, device accelerometer and gyroscope telemetry are polled to compute instantaneous pitch angles ϕ_tilt.



Fig. 1. End-to-end execution flow and decision pipeline of the NavEye framework. Multi-modal sensor inputs are processed via parallel edge neural networks; geometric depth and 3-corridor spatial reasoning with boundary hysteresis evaluate dynamic collision risk; high-priority directives preempt the central state machine to dispatch bilingual audio and haptic guidance.

B. Monocular Pinhole Geometric Distance and Obstacle Tracking

Rather than relying on heavy deep depth networks, NavEye implements a geometry-based distance estimation engine. Assuming standard perspective camera projection, the real-world distance Di to obstacle bi is calculated via object height prior H_real(ci) and effective bounding box height hi:

$$D_i = (f_y \cdot H_{real}(c_i)) / (h_i \cdot H_{img} \cdot \cos(\phi_{tilt})) \quad (2)$$

where fy is camera vertical focal length in pixels, H_img is sensor height, and ϕ_tilt compensates for handheld phone downward tilt. To avoid intermittent detection dropouts caused by lighting shadows or partial occlusions, a temporal centroid tracker maintains track state over a 2,000 ms grace window. Bounding boxes across consecutive frames are matched using Euclidean centroid gating: dist(c_t, c_{t-1}) < r_gate. Unmatched tracks decay gradually, preventing false negative collision hazards.

C. 3-Corridor Spatial Partitioning and Boundary Hysteresis

Assigning detected obstacles to lateral corridors allows NavEye to compute clear walking vectors. However, when an obstacle resides near corridor boundaries (e.g., x ≈ 0.35), minor sensor jitter or hand tremor induces rapid alternating classifications between Left and Center, causing the system to stutter conflicting steering commands ('Move left', 'Stop', 'Move left'). To guarantee state stability, NavEye introduces a boundary hysteresis algorithm with guard band δ = 0.04:

$$corridor(b_i, t) = \{ Left: x_i < \theta_L - \delta \cdot \mathbb{I}(prev=C);\ Right: x_i > \theta_R + \delta \cdot \mathbb{I}(prev=C);\ C: otherwise \} \quad (3)$$

By enforcing this dynamic Schmitt-trigger threshold, an obstacle currently in the Center path must cross beyond 0.31 to transition to Left, and beyond 0.69 to transition to Right. A temporal debounce filter further mandates that an obstacle state must remain continuous for ≥ 350 ms before initiating an evasive steering notification.

D. ArcFace Metric Learning and Identity Verification

Social navigation requires identifying familiar associates (e.g., family members, guides). NavEye integrates Google ML Kit for face landmark localization, followed by affine rotational alignment based on inter-ocular vector angles. Aligned 112×112 crops are passed to MobileFaceNet, producing a 512-dimensional normalized embedding vector e_t. Metric matching against enrolled local profiles (stored securely in SQLite `naveye.db`) utilizes a dual-component metric: centroid similarity S_cent and top-3 reference similarity S_top3. A verification match is confirmed if composite certainty exceeds threshold τ_face = 0.70.

E. Centralized Priority Preemption and State Arbitration

A critical vulnerability of naive assistive tools is race conditions between different sensory notifications (e.g., announcing a distant friend while the user is about to collide with a low wall). NavEye resolves this through an absolute priority preemption hierarchy:

- **Priority 1 (Emergency Collision Stop):** Distance D < 0.8 m in Center corridor. Immediately preempts ongoing audio with 0 ms queue latency, vibrating haptic actuators and commanding immediate stop.
- **Priority 2 (Evasive Steering Maneuver):** Obstacle within 0.8–2.0 m in Center corridor. Evaluates lateral corridor occupancy and commands deterministic steering ('Obstacle ahead, steer left').
- **Priority 3 ('Never Silence' Heartbeat):** If the central path remains unobstructed for 3,500 ms, the system issues a subtle reassurance tone or phrase ('Path clear, continue forward').
- **Priority 4 (Ambient Context & Currency):** Announces verified familiar identities, Indian banknotes (₹10–₹500), and OCR street text when no collision threats exist.

F. Bilingual Directive Synthesizer and Audio Output

Speech directives follow a deterministic 3-part natural grammar: [Spatial Direction] + [Distance/Obstacle] + [Action]. To eliminate language barriers in Tamil Nadu, all alerts are natively synthesized in Tamil (`ta-IN`) and English (`en-US`). Crucially, speech rate is user-adjustable between 1.0× and 2.0× to match experienced blind users' rapid auditory parsing habits.

V. SYSTEM IMPLEMENTATION

NavEye is developed using Flutter 3.x and Dart, with performance-critical neural inference executed in C++ via ONNX Runtime Mobile and Android NDK. The system runs entirely on-device without cloud dependency, ensuring total user privacy. Table I details the architectural specifications

and observed component latencies on a mid-range Qualcomm Snapdragon 778G test device. The comprehensive real-time guidance control loop is formalized in Algorithm 1.

Algorithm 1 NavEye Real-Time Multi-Modal Perception Loop

```
Require: Video stream F_t, IMU pitch ϕ_t, Priors H,
Guard δ, Thresholds D_crit, D_warn
1: for each frame F_t from CameraX do
2:     B_t ← INFER_YOLOv8n(F_t);   M_t ← NMS(B_t,
IoU=0.45, Conf=0.35)
3:     for each obstacle b_i ∈ M_t do
4:         D_i ← ESTIMATE_DEPTH(b_i.h, H(b_i.c), ϕ_t)
▷ Eq. (2)
5:         corr_i ← EVALUATE_HYSTERESIS(b_i.x,
prev_corr, δ)      ▷ Eq. (3)
6:         UPDATE_TRACKER(b_i, D_i, corr_i,
grace=2000ms)
7:     end for
8:     if ∃ b_k with corr_k == Center and D_k < D_crit
then
9:         DISPATCH_ALERT(Priority1, 'STOP!',
Haptics=Pulse)     ▷ Preemption
10:    else if ∃ b_k with corr_k == Center and D_k <
D_warn then
11:        dir ← SELECT_CLEAR_CORRIDOR(Left_occ,
Right_occ)
12:        DISPATCH_ALERT(Priority2, dir,
Bilingual=True)
13:    else if TIME_SINCE_LAST_ALERT() > 3500ms then
14:        DISPATCH_ALERT(Priority3, 'Path clear',
Mode=Heartbeat)
15:    end if
16: end for
```

TABLE I
SYSTEM SPECIFICATIONS AND EMBEDDED INFERENCE BENCHMARKS

Subsystem Component	Underlying Architecture	Latency / Runtime
Mobile Host OS	Android 14 (Flutter 3.x + C++ NDK)	Native Runtime
Object Detection	YOLOv8n (ONNX Runtime, INT8)	28.4 ms (35.2 FPS)
Face Verification	MobileFaceNet (ArcFace, INT8)	12.1 ms (Parallel)
Depth & Spatial Engine	Monocular Pinhole + Hysteresis	1.8 ms
Speech & Haptic Alert	Flutter TTS (Bilingual ta-IN/en-US)	3.4 ms
End-to-End Latency	Autonomous Edge Perception Loop	41.6 ms (24.0 FPS)

VI. EXPERIMENTAL EVALUATION

A. Dataset and Experimental Setup

To evaluate NavEye under demanding real-world conditions, we constructed a multi-scenario evaluation protocol comprising 500 navigation trials conducted across four distinct environmental categories: (i) narrow indoor corridors with doorways, (ii) cluttered classrooms with movable desks and chairs, (iii) outdoor sidewalks featuring vendor carts and pedestrians, and (iv) low-light stairwells (15–1,200 lx). Ground-truth distance and collision vectors were tracked using an external calibrated LiDAR baseline.

B. Baseline Systems for Comparison

NavEye was benchmarked against four representative mobility paradigms: (a) White Cane Baseline, (b) Cloud Vision Assistance (commercial app using cloud API over 4G LTE), (c) Standard Edge SSD (MobileNet-SSD without corridor hysteresis), and (d) NavEye without Preemption.

C. Performance Metrics

Performance is quantified across five core metrics: (1) Collision Avoidance Rate (CAR %), percentage of trials completed without obstacle contact; (2) False Alarm Rate (FAR %), alerts issued when path was clear; (3) End-to-End Guidance Latency (ms); (4) Steering Compliance Rate (SCR %), user successful adherence to steering directives; and (5) Cognitive Workload assessed via NASA-TLX index (0–100).

TABLE II
COMPARATIVE BENCHMARK WITH BASELINE SYSTEMS (%)

System	CAR	FAR	Lat (ms)	SCR	TLX
White Cane Only	61.4	4.2	N/A	58.2	68.4
Cloud Vision API	73.8	18.5	1640	69.1	54.2
Edge SSD (No Hyst)	82.5	21.3	74.2	76.4	49.8
NavEye (No Preempt)	89.6	7.8	43.1	86.5	38.6
NavEye (Full Proposed)	**95.8**	**3.6**	**41.6**	**94.2**	**26.4**

TABLE III
EFFECT OF BOUNDARY HYSTERESIS δ (ABLATION STUDY)

δ (guard)	Grace (ms)	CAR (%)	Flips/min	Delay (ms)
0.00	0	83.2	18.4	41.2
0.02	500	88.7	8.2	41.8
0.04 (Ours)	**2000 (Ours)**	**95.8**	**0.6**	**42.4**
0.06	3000	94.1	0.2	58.6
0.08	4000	91.3	0.1	74.2

D. Comparative Results and Ablation Findings

As reported in Table II, the full NavEye pipeline achieves a remarkable 95.8% collision avoidance rate, outperforming the white cane (61.4%) and cloud vision (73.8%). Cloud vision suffered catastrophic failures when cellular signal degraded in stairwells, causing latency spikes exceeding 3.5 seconds. NavEye's deterministic 41.6 ms latency ensured users received timely warnings before entering hazardous zones. Table III highlights the necessity of boundary hysteresis δ: without hysteresis (δ = 0.00), sensor jitter produced 18.4 direction flips per minute, disorienting users. Setting δ = 0.04 reduced steering oscillations to 0.6 flips/min while maintaining optimal latency.

E. User Study with Visually Impaired Participants

A formal usability trial was conducted with 12 visually impaired participants navigating an unfamiliar 45-meter indoor obstacle maze. Participants navigated with white cane alone, and subsequently with NavEye. NavEye reduced average navigation collisions from 4.8 to 0.4 per trial while increasing average walking velocity by 28.5%. Qualitative

feedback highlighted the invaluable confidence provided by the 'Never Silence' heartbeat and the clarity of native Tamil directives.

VII. DISCUSSION

A. Why On-Device Edge Processing Outperforms Cloud

For safety-critical blind navigation, millisecond determinism is non-negotiable. At typical walking speeds, an obstacle appearing around a corner requires reaction times under 200 ms. Cloud systems cannot guarantee this bounded response time due to wireless jitter, edge handoffs, and server queues. Furthermore, streaming live video of private domestic or office spaces to third-party cloud servers creates acute privacy and regulatory violations. NavEye proves that high-precision perception and spatial guidance can operate 100% locally on standard mobile chipsets, providing complete data sovereignty and zero operational network costs.

B. Failure Modes and Environmental Limitations

Empirical stress testing revealed three primary operational limitations: (i) transparent glass doors and floor-to-ceiling windows lack visual texture and are missed by RGB models (mitigated in NavEye by combining white cane tactile ground contact); (ii) rapid lighting transitions (e.g., exiting dark corridors into blinding sunlight) momentarily saturate camera sensors for 150–300 ms; and (iii) fast oncoming electric bicycles moving > 35 km/h compress available stopping distances below safe threshold margins.

C. Ethical, Social, and Safety Considerations

NavEye is explicitly designed as a complementary electronic travel aid that works in synergy with—rather than replacing—the white cane or guide dog. Auditory alerts are calibrated to preserve the user's natural acoustic hearing, and open-ear bone-conduction headphones are strictly recommended during outdoor use. Biometric face enrollment is restricted to explicit user consent, with embeddings stored exclusively in encrypted on-device SQLite tables without cloud backup.

VIII. CONCLUSION AND FUTURE WORK

We presented NavEye, a comprehensive edge-native, real-time multi-modal assistive navigation framework for blind and visually impaired individuals. By combining INT8-quantized YOLOv8n object detection, perspective-compensated monocular pinhole depth estimation, and a 3-corridor spatial reasoning engine with boundary hysteresis ($\delta = 0.04$), NavEye eliminates steering jitter while delivering deterministic 41.6 ms guidance. The framework enforces an absolute priority preemption hierarchy with a 'Never Silence' safety heartbeat and localized bilingual voice directives in Tamil and English. Empirical trials over 500 scenarios demonstrated 95.8% collision avoidance and a 61.4% reduction in cognitive fatigue.

Future research trajectories include: (i) incorporating spatial audio beamforming via binaural bone-conduction headphones to communicate obstacle azimuth directly through 3D acoustics; (ii) extending the vision engine with lightweight on-device stereoscopic and LiDAR fusion for depth mapping of transparent glass barriers; (iii) deploying fine-tuned on-device Small Language Models (SLMs) for complex conversational indoor scene navigation; and (iv) conducting longitudinal outdoor field studies across varied demographic cohorts.

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